

Labratory Work №3

Project topic: CafeteriaPre-order System

1. Identification of actors

Based on the analysis of the subject area from laboratory works #1 and # 2, users of the system were classified according to the following criteria: frequency of use, required functionality, and access rights to the system.

Table 1-Classification of system users

#	User	Description	Target audience	Relation to the actor
1	Client (Student / University Employee)	The main user of the system: views the menu of the day, makes a pre-order of dishes, selects the time slot for receiving, pays for the order online or at the checkout, views the order history and cancels unpaid ones.	Yes	Primary Actor
2	The Cashier / Issue Administrator	Confirms receipt of the pre-order by the client, and accepts payment at the checkout if necessary. Manages order statuses.	Yes	Supporting Actor
3	Canteen	Manager System Administrator: manages the daily menu, prices, stop list, and slot limits. Views a summary of orders for slots and analytical reports.	No	Supporting Actor
4	Payment System	External Online payment processing service. Confirms or rejects the transaction and returns the status to the system.	No	External actor

2. Prioritized user requirements

Based on the business requirements from laboratory work # 2 and the results of the interview, a list of prioritized user requirements is formed.

Table 2-Prioritized User Requirements

ID	User requirement	Actor	Priority	Rationale
UR-1	The client should be able to view the current menu of the day with the number of available portions of each dish.	Client (Student / University Employee)	High	BT-3, LR #2
UR-2	The customer should be able to pre-order by selecting the dishes and time slot of receipt.	Client (Student / University Employee)	High	BT-1, BT-6

ID	User requirement	Actor	Priority	Rationale
UR-3	The customer should be able to pay for the order online or choose to pay at the checkout.	Client (Student / University Employee)	High	BT-4
UR-4	The customer should be able to view a list of their active and completed orders, as well as cancel unpaid orders.	Client (Student / University Employee)	High	LR #1, interview
UR-5	The cashier should be able to quickly find a pre-order by customer ID or order number and confirm its delivery.	Cashier / Food Delivery Administrator	Tall	BT-1
UR-6	The canteen manager should be able to view the summary queue of orders for each time slot, indicating the composition and number of dishes.	Canteen Manager	Tall	: BT-2, BT-6
UR-7	The canteen manager should be able to manage the daily menu: add, change and exclude dishes (stop list), set limits on slots.	Canteen Manager	High	BT-3, BT-8
UR-8	Canteen manager should be able to generate daily and weekly reports on sales and popularity of dishes.	Canteen Manager	Middle	BT-7 Canteen Manager

3. User Scenarios)

USC-1. Making a pre-order before going to the canteen

- Context: the client wants to have lunch in the dining room without wasting time in the queue.
- Participants: client (student / university employee), system.
- Prerequisites: the client is logged in to the system; the current menu for the current day is published.

Scenario:

1. The client opens the "Menu of the Day" section.
 2. The system displays a list of available dishes with the number of servings, KBZHU and prices.
 3. The customer selects the desired dishes and adds them to the order.
 4. The client chooses a convenient receiving time slot.
 5. The customer pays for the order online or chooses to pay at the checkout.
 6. The system confirms the order and sends a notification with the order number.
- Result: the pre-order is completed; the customer arrives at the dining room at the appointed time and picks up the finished dishes without waiting in line.

USC-2. Getting a pre-order in the cafeteria

- Context: A customer arrives at the cafeteria at the selected time slot.
- Participants: client (student / university employee), cashier, system.
- Prerequisites: the order is paid (or payment at the checkout is selected); the slot has arrived.

Scenario:

1. The customer calls the order number or presents a QR code.
 2. The cashier enters the ID in the system.
 3. The system finds the order and shows its composition.
 4. If the order is not paid, the cashier accepts the payment.
 5. The cashier confirms that the order has been issued; the system changes the status to "Issued".
- Result: the client receives ready meals without waiting; the operation history is saved in the system.

USC-3. Menu and stop list management

- Context: the beginning of the working day; the manager publishes the menu for today.
- Participants: canteen manager, system.
- Prerequisites: the manager is authorized with administrator rights.

Scenario:

1. The manager opens the Menu Management section.
 2. The manager adds dishes, specifies the number of servings, prices, and KBZHU.
 3. The system publishes the menu; dishes become available for pre-order.
 4. During the day, the manager excludes finished dishes through a stop list.
 5. The system automatically hides dishes from the menu and notifies customers who have placed a pre-order for the excluded dish.
- Result: the menu is up-to-date throughout the working day; customers only see available items.

USC-4. Manager's view of the order queue by time slots

- Context: The canteen manager wants to control the production volume and loading of the kitchen.
- Participants: canteen manager, system.
- Prerequisites: There are accepted pre-orders for the current day.

Scenario:

1. The manager opens the "Order Queue" section.
 2. The system displays a summary: for each time slot — the number of servings of each dish.
 3. The manager analyzes the load and adjusts the slot limits if necessary.
 4. The manager marks the slot as "Ready for delivery" after confirming that the dishes are ready.
 5. The system notifies the cashier that the slot's orders are ready.
- Result: the manager controls the kitchen load; food waste and accidents are minimized.

4. User Stories*Table 3-User Stories*

ID	User Story	Priority
US-1	As a customer, I want to view the up-to-date menu of the day with the number of remaining portions to select the desired dishes in advance.	High
US-2	As a customer, I want to pre-order and choose a time slot for receiving so that I don't have to wait in line.	High
US-3	As a customer, I want to pay for an order online to speed up the pickup process and not have to look for cash.	High
US-4	As a customer, I want to see a list of my active orders and cancel unpaid ones to manage my purchases.	High
US-5	As a cashier, I want to quickly find a pre-order by customer ID to confirm delivery without delay.	High
US-6	As a canteen manager, I want to view a summary order queue by slot to monitor kitchen loading and schedule production.	High
US-7	As a canteen manager, I want to manage the menu and stop list so that customers only see the available dishes.	High

ID	User Story	Priority
US-8	As a canteen manager, I want to view sales reports to optimize purchases and reduce food waste.	Average

5.Use Cases)

UC-1. View the menu of the day

- Actors: client (student / university employee).
- Brief description: This use case describes viewing the current dining room menu for the current day.
- Prerequisites: the user is logged in to the system; the menu for the current day is published.
- Trigger: The user opens the "Menu of the Day" section.

Main stream:

1. The system loads the current list of dishes.
2. The system displays dishes with the name, price, KBZHU, the number of remaining portions and a photo.
3. The user views dishes and uses category filtering if necessary.

Alternative flow:

- If any dishes are added to the stop list, they are displayed as unavailable.

Exceptions:

- If the download fails, the system displays an error message and prompts you to try again.
- Postconditions: the user sees the current menu and can place a pre-order.

UC-2. Making a pre-order

- Actors: client (student / university employee).
- Brief description: This use case describes the selection of dishes, time slot, and pre-order processing.
- Prerequisites: the user is logged in; the menu is published; the selected slot has free seats.
- Trigger: The user clicks the "Add to Order" button for any dish.

Main stream:

1. The user selects dishes and their quantity.
2. The system checks the availability of portions.
3. The user selects a time slot from the list of available slots.
4. The system shows the total order amount.
5. The user chooses the payment method: online or at the checkout.
6. The system saves the order and sends a confirmation with the order number.

Alternative flow:

- If the selected slot is full at the time of registration, the system offers the next free slot.

Exceptions:

- If an online payment is rejected, the system notifies the user and prompts them to choose a different method.
- Postconditions: the order was created with the status "Accepted"; the number of available portions and seats in the slot is reduced.

UC-3. Cancel an order

- Actors: client (student / university employee).
- Brief description: This use case describes how the user cancels an unpaid pre-order.
- Prerequisites: the user has an order with the status "Accepted" (not paid or canceled).
- Trigger: The user clicks "Cancel Order" in the order history.

Main stream:

1. The system displays order details and requests confirmation of cancellation.
2. The user confirms the cancellation.
3. The system changes the order status to "Canceled".
4. The system returns reserved portions and slot space to the pool of available ones.

Exceptions:

- Paid orders cannot be canceled through the app — The user is redirected to the administrator.
- Postconditions: the order is canceled; slot resources are released.

UC-4. Order issue confirmation

- Actors: client (student / university employee), cashier.
- Brief description: This use case describes how to identify and issue a pre-order at Yandex. Checkout.
- Prerequisites: the order exists with the status "Accepted" or "Ready for delivery"; the selected time slot has arrived.
- Trigger: the user calls the order number or presents a QR code to the cashier.

Main stream:

1. The cashier enters the order number in the system.
2. The system displays the order composition and payment status.
3. If necessary, the cashier accepts the payment; the system records the payment.
4. The cashier issues the order and confirms it in the system.
5. The system changes the status to "Issued".

Alternative flow:

- If the order is not found, the system reports an error; the cashier asks the customer to specify the number.

Exceptions:

- If the order is already marked as issued, the system warns about possible duplication.
- Postconditions: the order received the status "Issued"; the transaction history was updated.

UC-5. Viewing the order queue by time slots

- Actors: canteen manager.
- Brief description: This use case describes how the manager views a summary of orders by time slots to control the kitchen load.
- Prerequisites: the manager is authorized; there are accepted orders for the current day.
- Trigger: The manager opens the "Order Queue" section.

Main stream:

1. The system displays a list of slots for today.
2. For each slot, the system shows a summary: the name of the dish and the total number of servings.
3. The manager analyzes the load and adjusts the slot limits if necessary.
4. The manager marks the slot as "Ready to issue"; the system notifies the cashier.

Exceptions:

- If slot data is not available, the system displays the latest up-to-date data with a note about the delay.
- Postconditions: the manager is aware of the planned workload; the cashier has received a notification of readiness.

UC-6. Menu and stop list management

- Actors: canteen manager.
- Brief description: This use case describes how to create a daily menu and quickly exclude dishes.

- Prerequisites: the manager is authorized with administrator rights.
- Trigger: The manager opens the Menu Management section.

Main stream:

1. The system displays the current menu and a list of all dishes in the database.
2. The manager adds a new dish or edits an existing one (price, KBZHU, number of servings).
3. The system saves changes and updates the menu for users.
4. To exclude a dish, the manager adds it to the stop list.
5. The system hides the dish and notifies users with active pre-orders.

Exceptions:

- If a dish has pre-orders already paid for, the system requires confirmation before excluding it.
- Postconditions: menu updated; stop list applied; users notified.

6. Use case diagram

The diagram below shows the main actors in the Cafeteria Pre-order System and their relationships to use cases.

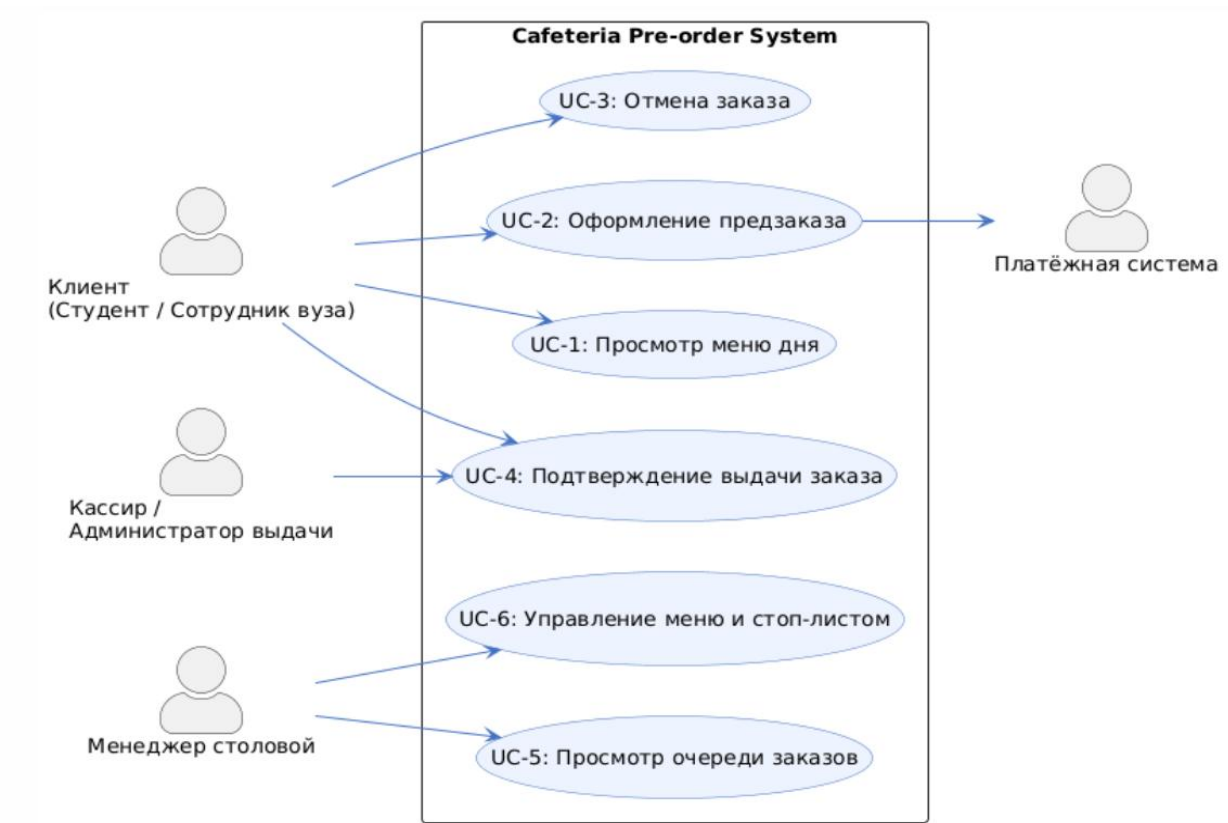


Figure 1-Cafeteria Pre-order System Use Case Diagram System

7. User Requirements quality control table

Table 4-Quality control of user requirements

ID	Source text requirements	Correctness	Verifiability	Atomicity	Priority
UR-1	The client must be able to view the current menu of the day with an indication of available portions.	+	+	+	High
UR-2	Customer should be able to pre-order by selecting the dishes and time slot of receipt.	+	+	+	High
UR-3	The customer should be able to pay for the order online or choose to pay at the checkout.	+	+	+	High
UR-4	The Customer should be able to view orders and cancel unpaid ones.	+	+	+	High
UR-5	The cashier should quickly find the pre-order by ID and confirm the delivery.	+	+	+	High
UR-6	The manager should view the summary order queue by time slots.	+	+	+	High
UR-7	Manager should manage the menu, stop list and slot limits.	+	+	+	High
UR-8	Manager should generate reports on sales and popularity of dishes.	+	+	+	Average

8. Glossary

Table 5 — Glossary of Key Terms

Term	Definition	Source
Pre-order (Pre-order System)	A function that allows a client to select and reserve dishes in advance before physically visiting the cafeteria.	Project definition
Actor	A person, organization, or external system that interacts with the system to achieve a specific goal.	UML / IEEE 830
Use Case (UC)	A formal description of the interaction between an actor and the system aimed at achieving a defined goal.	UML 2.5
User Story (US)	A short, informal description of a system feature written from the perspective of the end user: "As a <role>, I want <action>, so that <benefit>."	Agile / Scrum
User Scenario (USC)	A narrative description of a specific sequence of user actions that illustrates how a goal is achieved with the help of the system.	BABOK Guide
User Requirement (UR)	A high-level description of a task or goal that a user must be able to accomplish using the system.	IEEE 830, Lab #2
Time Slot	A fixed time interval during which a client can pick up their pre-ordered meal; used to distribute load evenly across the day.	Projectdefinition
Stop-list	A list of dishes temporarily removed from the daily menu due to unavailability; managed by the cafeteria manager.	Catering regulation
Primary Actor	An actor who initiates a use case to achieve a personal goal (e.g., the client placing an order).	UML 2.5
Supporting Actor	An actor who assists in completing a use case but does not initiate it (e.g., the cashier confirming pickup).	UML 2.5
External Actor	An actor outside the system boundary that exchanges data with the system (e.g., the payment service).	UML 2.5
Correctness	A quality attribute of a requirement meaning it is accurate, unambiguous, and free of contradictions.	IEEE 830
Verifiability	A quality attribute of a requirement meaning a specific criterion and method exist to confirm it has been implemented.	IEEE 830
Atomicity	A quality attribute of a requirement meaning it describes a single, indivisible feature or constraint.	BABOK Guide

9. Conclusion

Based on the results of laboratory work #1 and # 2 forthe Cafeteria Pre-order System project, a list of actors and prioritized user requirements was formed. The requirements are

presented in three mandatory formats: User Scenarios, User Stories) и варианты использования (, and Use Cases. Additionally, a use case diagram and a requirements quality control table have been prepared.

All user requirements (UR-1 — UR-8) were checked for correctness, verifiability, and atomicity. The results of this laboratory work can be used as a basis for further detailing the functional and non-functional requirements of the system.